

Petta lítur vel út. Förum í næsta skref, gera **Technical Specification v0.2 – Data Model**.

Frábært. Þá skulum við gera **Technical Specification v0.2 – Data Model** þannig að hún verði nægilega nákvæm til að nota síðar beint við PostgreSQL-hönnun og API-hönnun.

Ég myndi hafa eitt grundvallarmarkmið í huga:

Öll mikilvæg gögn í málinu eiga að vera rekjanleg aftur að Case og hægt að sjá hver stofnaði, breytti, skoðaði eða tengdi gögnin.

Technical Specification v0.2

Icelandic Legal Case Management System — Data Model

Version: 0.2

Status: Design Draft

Primary database: PostgreSQL

Character set: UTF-8

Primary language: Icelandic UI / English technical identifiers

Architecture: Relational core + object storage for files

1. Database design principles

1.1 Naming

Database identifiers verða á ensku:

cases

parties

documents

evidence

court_proceedings

en notendaviðmótið á íslensku:

Mál

Aðilar

Skjöl

Sönnunargögn

Dómsmál

Þetta gerir kerfið auðveldara að viðhalda og mögulegt að bæta við öðrum tungumálum síðar.

1.2 Primary keys

Ég myndi nota **UUID** fyrir primary keys:

id UUID PRIMARY KEY

Ekki:

id SERIAL

Ástæðan er meðal annars að UUID hentar betur ef kerfið verður síðar dreift yfir fleiri kerfi eða þarf að flytja gögn milli gagnagrunna.

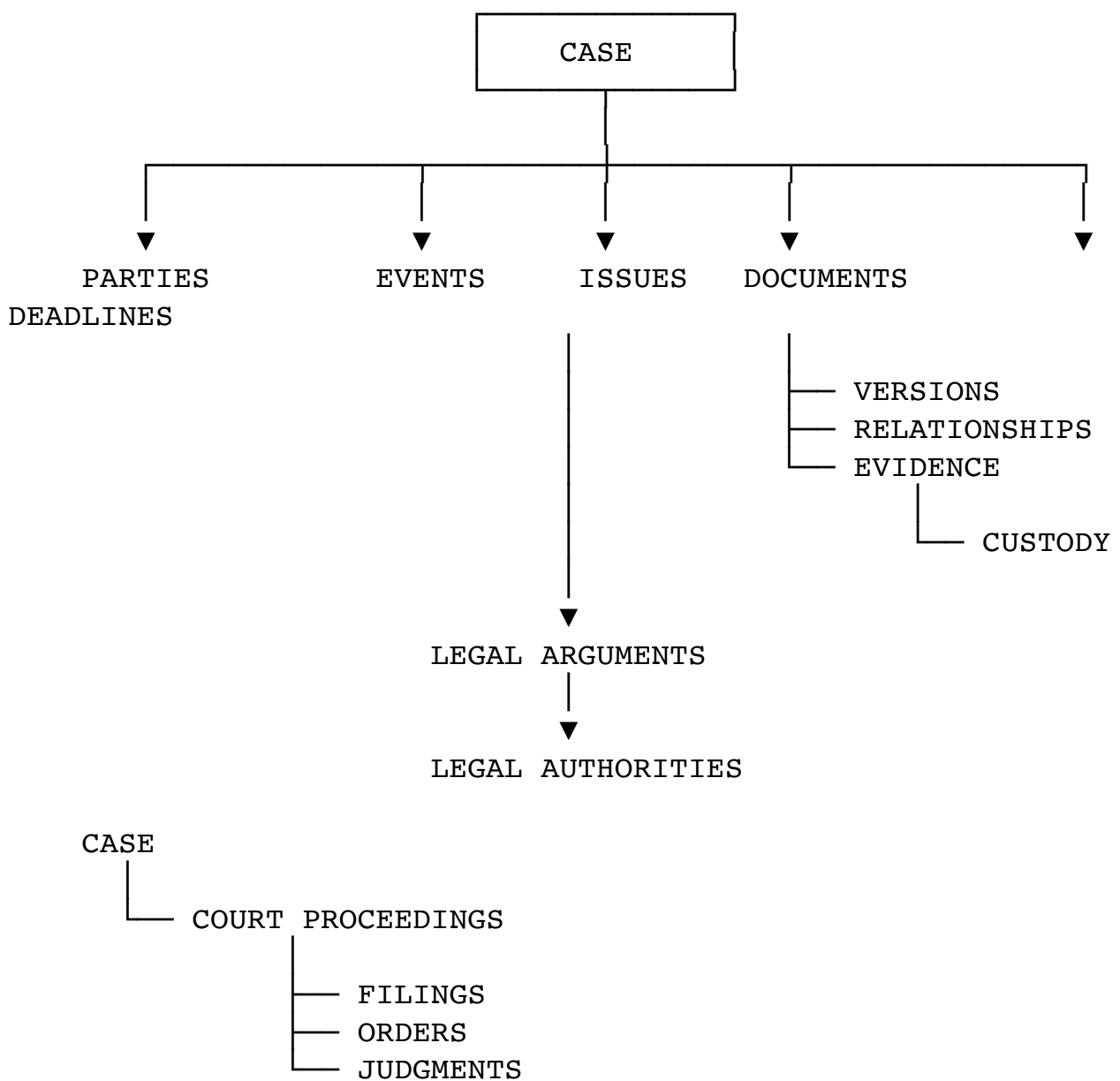
1.3 Timestamps

Allir timestamps skulu geymdir sem:

`TIMESTAMPZ`

og UTC notað sem canonical storage format.

2. Core entity relationship model



3. cases

Hjarta kerfisins.

Column	Type	Required	Description
id	UUID	✓	Primary key
case_number	VARCHAR(50)	✓	Innra málsnúmer
title	VARCHAR(500)	✓	Heiti máls
description	TEXT		Lýsing
case_type_id	UUID		Tegund máls
status	VARCHAR(50)	✓	Staða
priority	VARCHAR(20)		Forgangur
opened_at	TIMESTAMP Z	✓	Stofndagur
closed_at	TIMESTAMP Z		Lokadagur
responsible_user_id	UUID		Ábyrgðaraðili
confidentiality_level_id	UUID	✓	Trúnaðarstig
retention_policy_id	UUID		Varðveislustefna
created_by	UUID	✓	Stofnandi
created_at	TIMESTAMP Z	✓	
updated_at	TIMESTAMP Z	✓	

Status

NEW
OPEN
ON_HOLD
PRE_LITIGATION
LITIGATION
APPEAL
JUDGMENT
CLOSED
ARCHIVED

4. persons

Einstaklingar eru geymdir óháð málum.

persons

Column	Type
id	UUID
first_name	VARCHAR(200)
middle_name	VARCHAR(200)
last_name	VARCHAR(200)
display_name	VARCHAR(500)
nationality	VARCHAR(100)
email	VARCHAR(320)
phone	VARCHAR(50)
address	TEXT
created_at	TIMESTAMP Z
updated_at	TIMESTAMP Z

Ath: Ég myndi ekki setja kenntölu í þessa töflu í MVP nema raunveruleg krafa sé fyrir hendi. Ef hún verður nauðsynleg þarf hún sérstaka vernd og aðgangsstýringu.

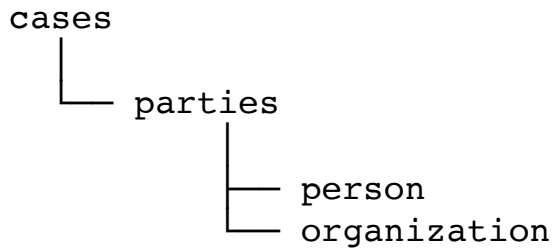
5. organizations

organizations

Column	Type
id	UUID
name	VARCHAR(500)
registration_number	VARCHAR(50)
organization_type	VARCHAR(100)
address	TEXT
email	VARCHAR(320)
phone	VARCHAR(50)
created_at	TIMESTAMP Z
updated_at	TIMESTAMP Z

6. parties

Petta er tengitafla milli aðila og máls.



Column	Type
id	UUID
case_id	UUID FK
person_id	UUID FK
organization_id	UUID FK
role	VARCHAR(50)
legal_representative_id	UUID
created_at	TIMESTAMPTZ

Constraint:

person_id XOR organization_id
þ.e. party má annaðhvort vera einstaklingur eða organization.

Party roles

CLAIMANT
DEFENDANT
APPELLANT
RESPONDENT
WITNESS
THIRD_PARTY
EXPERT
OTHER

7. events

Málsatvik.

events

Column	Type
id	UUID
case_id	UUID FK
event_date	DATE
event_time	TIME
title	VARCHAR(500)
description	TEXT
event_type	VARCHAR(100)
location	VARCHAR(500)
created_by	UUID
created_at	TIMESTAMP T Z

Event types

CONTRACT_SIGNED
 COMMUNICATION
 MEETING
 PAYMENT
 BREACH
 NOTICE
 COMPLAINT
 COURT_EVENT
 OTHER

8. documents

Þetta er eitt mikilvægasta table-ið.

documents

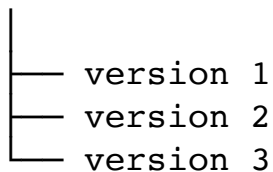
Column	Type
id	UUID
case_id	UUID FK
document_number	VARCHAR(50)
document_type_id	UUID
title	VARCHAR(500)
original_filename	VARCHAR(1000)

canonical_filename	VARCHAR(1000)
document_date	DATE
received_date	DATE
author	VARCHAR(500)
sender	VARCHAR(500)
recipient	VARCHAR(500)
source	VARCHAR(500)
mime_type	VARCHAR(200)
file_size	BIGINT
ocr_status	VARCHAR(50)
confidentiality_level_id	UUID
legal_privilege	BOOLEAN
current_version_id	UUID
created_by	UUID
created_at	TIMESTAMPTZ
updated_at	TIMESTAMPTZ

9. document_versions

Aldrei yfirstkrifa document binary.

documents



document_versions

Column	Type
id	UUID
document_id	UUID FK
version_number	INTEGER
object_storage_key	VARCHAR(2000)
sha256	CHAR(64)
file_size	BIGINT
mime_type	VARCHAR(200)
is_original	BOOLEAN

created_by	UUID
created_at	TIMESTAMPTZ
change_reason	TEXT

Constraint:

UNIQUE(document_id, version_number)

10. document_relationships

Gerir kleift að tengja skjöl.

document_relationships

Column	Type
id	UUID
source_document_id	UUID
target_document_id	UUID
relationship_type	VARCHAR(50)
created_by	UUID
created_at	TIMESTAMPTZ

Relationship types:

VERSION_OF
 COPY_OF
 DERIVED_FROM
 TRANSLATION_OF
 SUMMARY_OF
 RESPONDS_TO
 REFERENCES
 ATTACHMENT_OF
 REDACTION_OF

11. evidence

Sönnunargagn er **notkun/hlutverk skjals í máli**, ekki endilega sérstakt file.

evidence

Column	Type
id	UUID

case_id	UUID
document_id	UUID
evidence_type	VARCHAR(100)
description	TEXT
source	VARCHAR(500)
obtained_at	TIMESTAMPTZ
obtained_by	UUID
custodian	VARCHAR(500)
original_format	VARCHAR(100)
integrity_status	VARCHAR(50)
relevance	VARCHAR(50)
admissibility_statuses	VARCHAR(50)
created_at	TIMESTAMPTZ

12. evidence_custody_events

evidence_custody_events

Column	Type
id	UUID
evidence_id	UUID
timestamp	TIMESTAMPTZ
action	VARCHAR(100)
from_user_id	UUID
to_user_id	UUID
location	VARCHAR(500)
reason	TEXT
sha256	CHAR(64)
notes	TEXT
created_by	UUID

Actions

RECEIVED
IMPORTED
TRANSFERRED
COPIED
EXPORTED
RETURNED
ARCHIVED

13. issues

Ágreiningsefni.

issues

Column	Type
id	UUID
case_id	UUID
title	VARCHAR(500)
description	TEXT
status	VARCHAR(50)
priority	VARCHAR(20)
created_by	UUID
created_at	TIMESTAMPTZ
resolved_at	TIMESTAMPTZ

14. facts

Mjög gagnleg eining.

facts

Column	Type
id	UUID
case_id	UUID
statement	TEXT
status	VARCHAR(50)

importance	VARCHAR(20)
created_by	UUID
created_at	TIMESTAMP Z

Status

ALLEGED
 ADMITTED
 DISPUTED
 PROVEN
 DISPROVEN
 UNKNOWN

15. legal_arguments

legal_arguments

Column	Type
id	UUID
case_id	UUID
issue_id	UUID
title	VARCHAR(500)
argument_text	TEXT
position	VARCHAR(50)
status	VARCHAR(50)
created_by	UUID
created_at	TIMESTAMP Z
updated_at	TIMESTAMP Z

16. Tensingar evidence → facts → arguments

Petta þarf að vera many-to-many.

fact_evidence

 fact_id

evidence_id

og:

argument_facts

argument_id

fact_id

og:

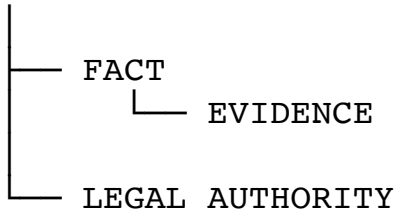
argument_authorities

argument_id

authority_id

Þannig getum við fengið:

LEGAL ARGUMENT



Þetta er ein mikilvægasta hönnunarákvörðunin í kerfinu.

17. legal_authorities

legal_authorities

Column	Type
id	UUID
authority_type	VARCHAR(50)
title	VARCHAR(1000)
citation	VARCHAR(1000)
date	DATE
court	VARCHAR(500)
reference	VARCHAR(1000)
url	VARCHAR(2000)
summary	TEXT
created_at	TIMESTAMPTZ

Types:

LAW
REGULATION
COURT_JUDGMENT
COURT_ORDER
EU_LEGISLATION
EU_JUDGMENT
LEGAL_COMMENTARY
OTHER

18. court_proceedings

Eitt Case getur haft fleiri en eitt dómsmál.

court_proceedings

Column	Type
id	UUID
case_id	UUID
court	VARCHAR(500)
court_case_number	VARCHAR(100)
court_level	VARCHAR(100)
jurisdiction	VARCHAR(200)
filing_date	DATE
status	VARCHAR(50)
judge	VARCHAR(500)
created_at	TIMESTAMPZ

Dæmi:

CASE-001
├── Héraðsdómur
└── Landsréttur

19. court_filings

court_filings

Column	Type
id	UUID
proceeding_id	UUID
document_id	UUID
filing_type	VARCHAR(100)
filing_date	TIMESTAMP Z
submitted_by	UUID
submitted_to	VARCHAR(500)
submission_method	VARCHAR(100)
receipt_timestamp	TIMESTAMP Z
court_status	VARCHAR(100)

Status:

DRAFT

REVIEW

APPROVED

SUBMITTED

RECEIVED

FILED

REJECTED

WITHDRAWN

20. deadlines

deadlines

Column	Type
id	UUID
case_id	UUID
proceeding_id	UUID
title	VARCHAR(500)
description	TEXT
due_at	TIMESTAMP Z

responsible_user_id	UUID
status	VARCHAR(50)
source	VARCHAR(500)
reminder_at	TIMESTAMPTZ

Status:

OPEN

COMPLETED

OVERDUE

CANCELLED

21. legal_holds

legal_holds

Column	Type
id	UUID
case_id	UUID
reason	TEXT
status	VARCHAR(30)
created_by	UUID
created_at	TIMESTAMPTZ
released_by	UUID
released_at	TIMESTAMPTZ

Critical constraint:

ACTIVE LEGAL HOLD

↓

NO AUTOMATED DELETION

22. retention_policies

retention_policies

Column	Type
id	UUID

name	VARCHAR(500)
retention_period_days	INTEGER
trigger_event	VARCHAR(100)
approval_required	BOOLEAN
legal_hold_required	BOOLEAN
created_at	TIMESTAMP T Z

Ég myndi **ekki** setja íslenska lögbundna varðveislutíma beint í gagnalíkanið. Þeir eiga að vera configurable policies sem hægt er að uppfæra án þess að breyta database schema.

23. users

users

Column	Type
id	UUID
external_subject	VARCHAR(500)
username	VARCHAR(200)
display_name	VARCHAR(500)
email	VARCHAR(320)
status	VARCHAR(50)
created_at	TIMESTAMP T Z
last_login_at	TIMESTAMP T Z

`external_subject` gerir okkur kleift að nota:

- Microsoft Entra ID
- Keycloak
- Auth0
- annað OIDC provider

án þess að breyta innra user model.

24. Roles & permissions

roles

id

name

description

permissions

id

resource

action

Dæmi:

CASE	READ
------	------

CASE	WRITE
------	-------

DOCUMENT	READ
----------	------

DOCUMENT	CREATE
----------	--------

DOCUMENT	EXPORT
----------	--------

EVIDENCE	READ
----------	------

EVIDENCE	TRANSFER
----------	----------

CASE	CLOSE
------	-------

AUDIT	READ
-------	------

Og:

user_roles

user_id

role_id

25. Case-level permissions

RBAC eitt og sér er ekki nóg.

case_permissions

case_id

user_id

permission

Dæmi:

CASE-001

—	Lawyer A → READ/WRITE
—	Lawyer B → READ

└─ External Counsel → READ/EVIDENCE
└─ Admin → ADMIN

26. audit_log

Þetta verður **append-only**.

audit_log

Column	Type
id	UUID
timestamp	TIMESTAMPTZ
user_id	UUID
action	VARCHAR(100)
entity_type	VARCHAR(100)
entity_id	UUID
old_value	JSONB
new_value	JSONB
ip_address	INET
reason	TEXT

Actions:

CREATE
UPDATE
DELETE
VIEW
DOWNLOAD
EXPORT
LOGIN
LOGOUT
SHARE
TRANSFER
SUBMIT
APPROVE
REJECT

27. Metadata / configurable lookup tables

Ég myndi **ekki** nota PostgreSQL ENUM fyrir flesta þessa reiti.

Til dæmis:

document_type
case_type
event_type
confidentiality_level
evidence_type
ættu frekar að vera configurable tables:

document_types
case_types
event_types
confidentiality_levels
Þannig er hægt að bæta við:

FOREIGN_JUDGMENT
EXPERT_REPORT
FORENSIC_REPORT
án database migration.

28. Full-text search

PostgreSQL getur byrjað á:

tsvector
fyrir:

documents
facts
issues
legal_arguments

En ég myndi gera architecture þannig að síðar sé hægt að bæta við:

OpenSearch
fyrir:

- OCR text
- fuzzy search
- faceted search
- multilingual search
- semantic search
- vector search

29. File storage

Ekki geyma PDF/DOCX/MP4 binaries í PostgreSQL.

Database:

```
document_versions
└── object_storage_key
```

Object storage:

```
s3://legal-case-data/
    cases/
        CASE-2026-00123/
            documents/
                ...
```

Mikilvægt:

Object storage key skal ekki vera byggður eingöngu á filename.

Til dæmis:

```
cases/{case_uuid}/documents/{document_uuid}/{version_uuid}
```

30. Integrity model

Við innflutning:

```
FILE
├── SHA-256
├── MIME detection
├── Malware scan
└── Metadata extraction
```

SHA-256:

```
document_versions.sha256
```

Pannig:

```
original file
└── SHA256 = X
```

stored file

└─ SHA256 = X

Ef:

SHA256 != X

skal kerfið flagga integrity failure.

31. Mikilvæg business constraints

Ég myndi setja eftirfarandi sem **hard requirements**.

Document

A document cannot be permanently deleted

if:

legal_hold = ACTIVE

Evidence

Evidence cannot be altered.

Ef lýsing eða metadata breytist:

AUDIT_LOG

Original

Original document version cannot be modified.

Audit

Audit records cannot be modified through normal application APIs.

Case

Closed case cannot be modified
without explicit reopen permission.

32. GDPR / Privacy by design

Database-hönnunin ætti að fylgja:

Data minimisation

Purpose limitation

Access control

Encryption

Auditability

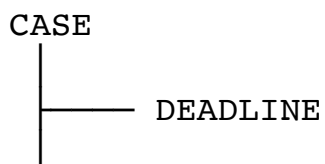
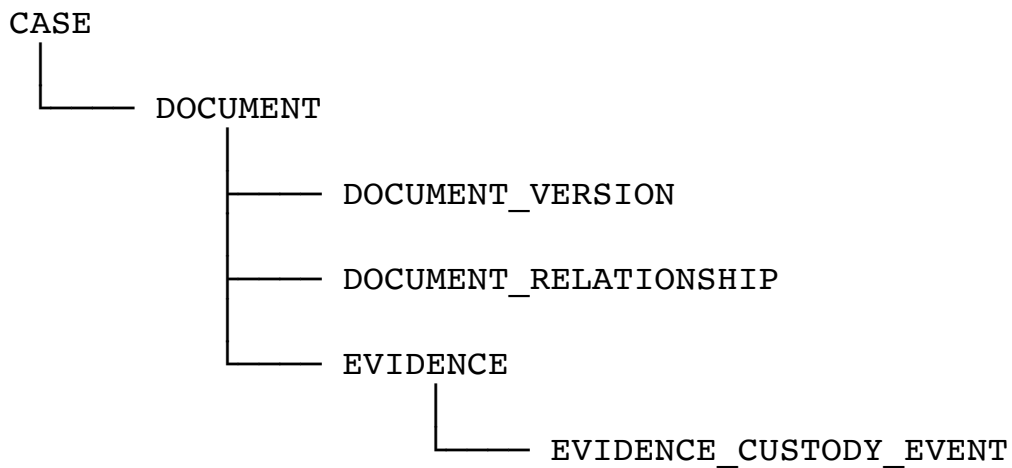
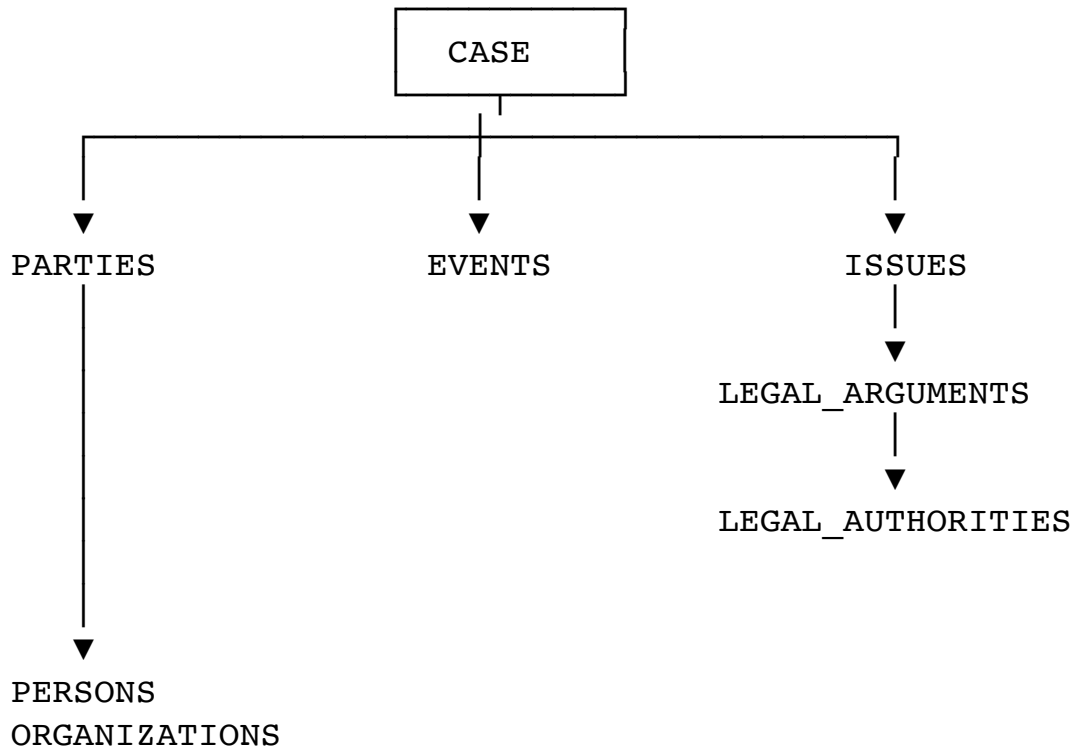
Retention

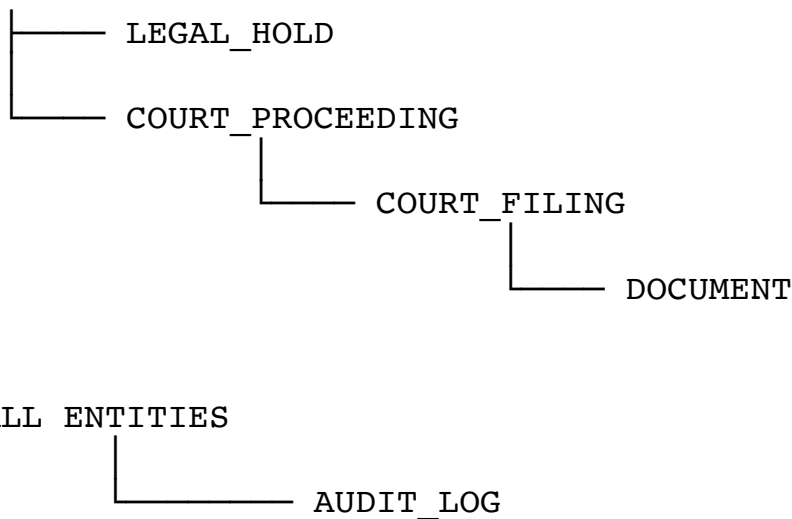
Deletion/anonymisation where legally permitted

En hér þarf að passa sérstaklega upp á að „**right to delete**“ sé ekki útfært sem einfalt **DELETE FROM**. Lagaleg varðveisla, sönnunargögn og legal hold geta haft forgang og þurfa sérfræðimat.

33. Logical ER model

Heildarmyndin verður þá:





34. Mikilvæg hönnunarákvörðun: Case-centric vs Document-centric

Ég mæli mjög eindregið með:

✗ Document-centric

Folder
└─ files

✓ Case-centric

Case
├─ Parties
├─ Events
├─ Issues
├─ Facts
├─ Documents
├─ Evidence
├─ Arguments
├─ Legal Authorities
├─ Deadlines
└─ Court Proceedings

Þetta er það sem gerir kerfið að **Legal Case Management System**, en ekki bara DMS.

35. MVP database

Ég myndi byrja á þessum töflum:

CORE

cases
persons
organizations
parties
users

DOCUMENT

documents
document_versions
document_relationships

CASE ANALYSIS

events
issues
facts
legal_arguments
legal_authorities

EVIDENCE

evidence
evidence_custody_events

COURT

court_proceedings
court_filings
deadlines

SECURITY

roles
permissions
user_roles
case_permissions
audit_log

RETENTION

legal_holds
retention_policies

Þetta eru **25 töflur**, en ekki allar þurfa að vera í fyrstu migration.

Næsta útgáfa: v0.3

Nú erum við komin á þann stað að næsta skref getur verið mjög hagnýtt:

Technical Specification v0.3 – PostgreSQL Schema & API

Ég myndi þá gera þrjá hluti saman:

A. PostgreSQL

```
CREATE TABLE cases (...);  
CREATE TABLE persons (...);  
CREATE TABLE documents (...);
```

...

með:

- PK/FK
- indexes
- unique constraints
- check constraints
- triggers
- audit mechanisms

B. ER diagram

Sýnilegt diagram sem sýnir allar 1 : N og M : N tengingar.

C. REST API specification

Til dæmis:

```
POST    /api/v1/cases  
GET     /api/v1/cases/{id}  
PATCH  /api/v1/cases/{id}  
  
POST    /api/v1/cases/{id}/documents  
GET     /api/v1/documents/{id}  
GET     /api/v1/documents/{id}/versions  
  
POST    /api/v1/cases/{id}/evidence  
POST    /api/v1/evidence/{id}/custody-events  
  
POST    /api/v1/cases/{id}/events  
POST    /api/v1/cases/{id}/issues
```

POST /api/v1/cases/{id}/arguments

POST /api/v1/proceedings

POST /api/v1/proceedings/{id}/filings

Þá erum við komin með **database + application interface specification**, sem er í raun næsta stig frá hönnun yfir í forritanlegt kerfi.